Mahavir Education Trust's
Shah & Anchor Kutchhi Engineering College
An Autonomous Institute Affiliated to University of Mumbai

Department of Cyber Security

Self-Learning Assessment Report (MOOC)

Student Information

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Stage 1: Activity 1 – Article Writing

Chosen Topic: CompTIA Cybersecurity Analyst+: Authorization

Article Summary (300–400 words):

Authorization is a critical concept in cybersecurity and a key focus area in the CompTIA Cybersecurity Analyst+ (CySA+) certification. It refers to the process of determining what actions an authenticated user or system is permitted to perform within a network or application. While authentication verifies identity, authorization defines access rights and privileges, ensuring users only interact with resources they are allowed to use.

In modern cybersecurity environments, authorization is typically enforced using access control models. The most common models include Role-Based Access Control (RBAC), Attribute-Based Access Control (ABAC), and Mandatory Access Control (MAC). RBAC assigns permissions based on user roles, making it efficient for organizations with defined job responsibilities. ABAC, on the other hand, uses attributes such as user behavior, location, and time to make dynamic access decisions. MAC is stricter and often used in government or military systems, where access is based on classification levels.A key principle associated with authorization is the Principle of Least Privilege (PoLP). This ensures users are granted the minimum level of access necessary to perform their tasks, reducing the risk of accidental or malicious misuse. Implementing least privilege helps minimize attack surfaces and limits the impact of compromised accounts.

Authorization mechanisms are enforced through tools like Access Control Lists (ACLs), Identity and Access Management (IAM) systems, and policy enforcement points. These systems continuously evaluate access requests and ensure

compliance with organizational security policies. Logging and monitoring authorization activities are also essential, as they help detect unauthorized access attempts and support incident response.

In the CySA+ context, understanding authorization is important for threat detection and response. Analysts must evaluate whether access anomalies—such as privilege escalation or unauthorized data access—indicate a security breach. Misconfigured authorization controls are a common vulnerability exploited by attackers, making proper implementation and regular auditing crucial.

In summary, authorization is a foundational security control that governs access to systems and data. Within the CySA+ framework, it plays a vital role in protecting organizational assets, enforcing policies, and enabling cybersecurity professionals to detect and respond to threats effectively.